

Anatomy-Aware Cross-Modal Medical Image Synthesis

Rethinking CT $\leftrightarrow$ MRI Translation under Imperfect Paired Supervision

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Roadmap

1 Motivation & Problem

Clinical asymmetry of CT & MRI; why “paired” data is not truly paired; re-framing the objective.

2 Data & Pipeline

SynthRAD 2023, patient-wise splits, 2.5D context, intensity normalisation.

3 Models (8 in a $2 \times 2 \times 2$ design)

Pix2Pix | CycleGAN | Paired Diffusion | Unpaired Cycle-Diffusion.

4 Quantitative Results

Test-set metrics, training curves, distribution analysis.

5 Ablation Study

Backbone $\times$ Pairing $\times$ Region; main effects & interactions.

6 Qualitative & Discussion

Brain & pelvis comparisons; perceptual $\leftrightarrow$ pixel trade-off; limitations & future work.

Clinical Motivation: Why CT↔MRI Translation?

The clinical asymmetry

- **CT**: electron-density (HU) for dose calculation, fast, ionising
- **MRI**: superior soft-tissue contrast, no radiation, slow & expensive
- Many radiotherapy and surgical-planning workflows want *both*

Why *synthetic* cross-modality is useful

- MR-only RT planning needs synthetic CT for dose
- CT-only triage centres benefit from synthetic MRI for soft-tissue review
- Reduced cost, scan-time and radiation burden

SynthRAD 2023 dataset

Property	Value
Regions	Brain, Pelvis
Modalities	CT, MRI (T1/T2)
Patients	180 (144 train / 36 test)
Split	Patient-wise (no leakage)
HU clip	[−1000, 1000]

Core challenge

Non-linear intensity mapping across a large domain gap, with imperfect anatomical correspondence even in “paired” scans.

Problem Statement: “Paired” Data Is Not Truly Paired

Sources of mis-pairing in clinical CT–MRI

- **Time gap** between acquisitions (hours–weeks)
- **Organ motion**: breathing, peristalsis, cardiac
- **Bladder & bowel filling** (pelvis)
- **Tumour progression** or post-treatment change
- **Patient repositioning**, scanner-bed bias
- **Fat / oedema drift**

Consequence

Pixel-perfect $G : X_{CT} \rightarrow X_{MR}$ is *ill-posed*: the target itself drifts between scans. Strict paired supervision can therefore *over-constrain* learning.

Re-framing the objective

*From “predict the exact ground-truth voxel” to “produce an **anatomically plausible** translation that is consistent with the source under cycle and perceptual constraints.”*

Research questions

- 1 How much does **pairing supervision** actually buy under imperfect alignment?
- 2 Does a **distilled latent-diffusion** prior beat GAN backbones at perceptual realism?
- 3 Are the resulting trade-offs **additive** or **interactive** across backbone $\times$ pairing?

Why Existing Methods Fall Short

Pix2Pix (Isola+ '17)

- Strong paired supervision (L1+adv)
- Sensitive to mis-alignment $\Rightarrow$ over-smoothed, mode-collapsed outputs
- Cannot exploit unpaired archives

CycleGAN (Zhu+ '17)

- Cycle-consistency, no paired data
- Weak structural constraints $\Rightarrow$ *anatomy hallucinations*
- Mode collapse & checkerboard artefacts in 2D

Vanilla DDPM

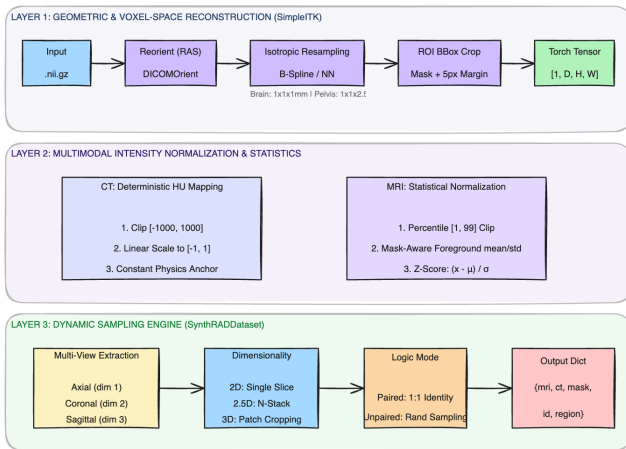
- Excellent generative prior
- ~ 1000 -step sampling: prohibitive for 3D medical volumes
- Lacks medical-domain conditioning out-of-the-box

Our contribution

We benchmark four paradigms $\times$ two regions = eight models, then run a research-grade ablation under *patient-wise* evaluation. We use SD-Turbo + LoRA as a one-step latent-diffusion backbone, evaluated against Pix2Pix and CycleGAN, in both *paired* and *cycle* settings.

Dataset & Engineering Pipeline

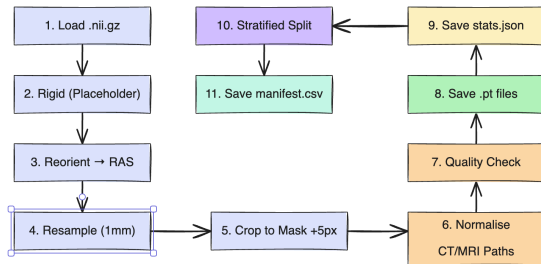
SynthRAD 2023: Data Engineering Pipeline



NIfTI ingestion → RAS reorient → resample → mask-guided crop → intensity normalisation → 2D / 2.5D slice sampling → augmentation → trainer-specific tensor cache.

Preprocessing Workflow & Splits

Preprocessing Workflow Details



Patient-wise split (*no leakage*)

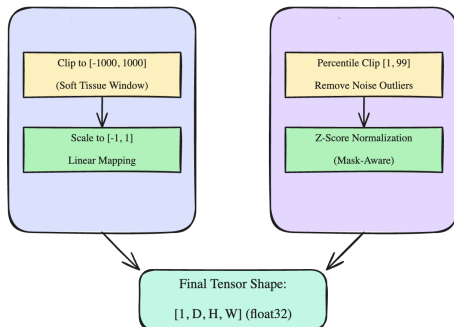
- 144 train / 36 test patients
- All slices of a patient are in the *same* fold
- Slice-wise splits inflate metrics by $\sim 15\text{--}30\%$ via memorisation

Resolution choice

GANs at 256^2 , diffusion at 512^2 to match SD-Turbo's pre-trained manifold. Higher resolution preserves small lesions.

Intensity Normalisation & 2.5D Sampling

Data Normalization Strategy



2.5D context

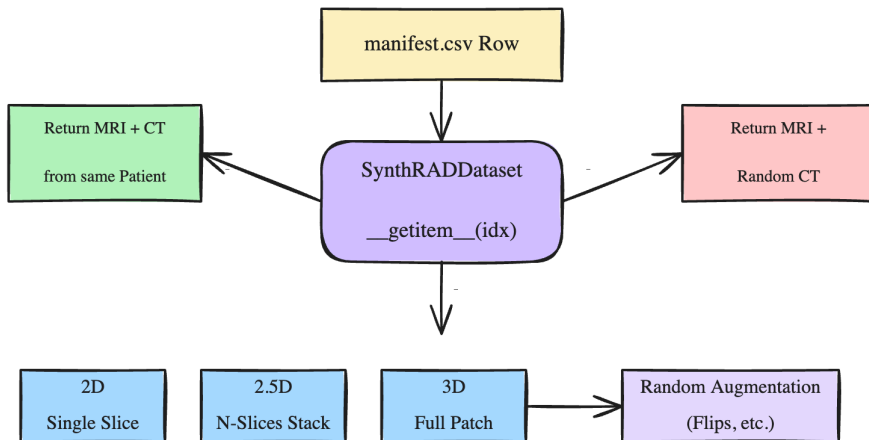
- 3 adjacent axial slices stacked into the channel dim
- Preserves through-plane anatomical coherence
- $\sim 3\times$ cheaper than full 3D, much closer to anatomical reality than pure 2D

Tissue windowing

- Brain CT: standard brain window before $[-1, 1]$ mapping
- Pelvis MRI: percentile clip $[1\%, 99\%]$ + foreground Z-score
- Soft-tissue windowing essential – otherwise anatomy invisible

Dataloader Logic

Dataset & DataLoader Logic



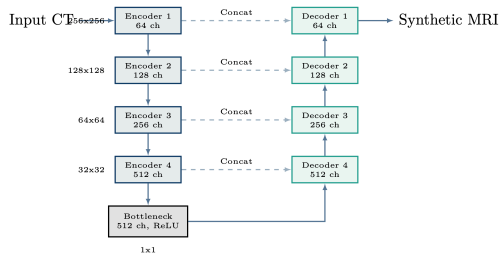
The 8-Model Factorial Design

Pairing	GAN backbone		Latent-Diffusion backbone	
	Brain	Pelvis	Brain	Pelvis
Paired	Pix2Pix-B	Pix2Pix-P	PairedDiff-B	PairedDiff-P
Unpaired	CycleGAN-B	CycleGAN-P	UnpairedDiff-B	UnpairedDiff-P

Why a fully-crossed design?

A balanced $2 \times 2 \times 2$ factorial (*backbone* $\times$ *pairing* $\times$ *region*) lets us decompose the variance into **main effects** (does pairing help on average?) and an **interaction** (does pairing help *diffusion* more than GAN?). Identical test-set slices across all four within-region cells enable paired non-parametric tests.

Pix2Pix — Paired GAN



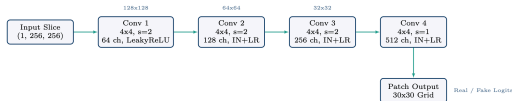
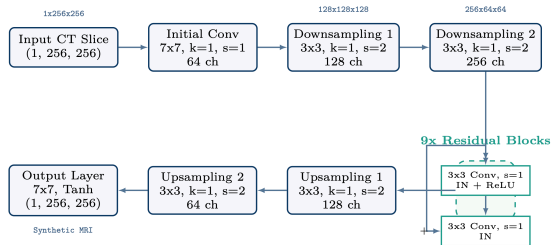
Discriminator: 70×70 PatchGAN on $\text{concat}(x, y)$.

Generator: U-Net (8-layer, skip connections), 4×4 stride conv, BN, LeakyReLU $\rightarrow$ Tanh.

Loss $\mathcal{L}_G = \mathcal{L}_{\text{LSGAN}} + \lambda_{L1} \|y - G(x)\|_1$, $\lambda_{L1} = 100$.

Adam ($\beta_1=0.5$), LR 2×10^{-4} , 500 epochs (250 const + 250 decay), batch 1, BatchNorm.

CycleGAN — Unpaired GAN

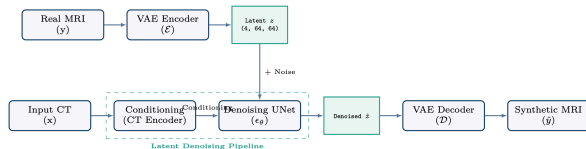


Multi-scale PatchGAN (2 scales, $2 \times$ avg-pool input).

Generator: ResNet-9, 7×7 reflection-pad, $2 \times$ down/up, optional self-attention.

Losses adversarial + $\lambda_{\text{cyc}} \cdot \mathcal{L}_{\text{cyc}} + \lambda_{\text{idt}} \cdot \mathcal{L}_{\text{idt}} + \lambda_{\text{LPIPS}} \cdot \mathcal{L}_{\text{LPIPS}}$, $\lambda_{\text{cyc}}=10$, $\lambda_{\text{idt}}=0.5$, $\lambda_{\text{LPIPS}}=1$.
Adam, LR 2×10^{-4} , 500 epochs, Instance-Norm.

Paired Latent Diffusion — SD-Turbo + LoRA



Configuration

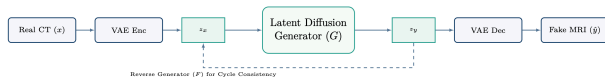
- Base: stabilityai/sd-turbo (1-step LDM)
- LoRA: UNet rank $r=8$, VAE rank $r=4$
- Resolution 512^2 , 2.5D input
- AdamW, LR 5×10^{-6} , grad-accum 8

Loss

$$\mathcal{L} = \lambda_d \mathcal{L}_{\text{denoise}} + \lambda_1 \mathcal{L}_{L1} + \lambda_P \mathcal{L}_{\text{LPIPS}} + \lambda_G \mathcal{L}_{\text{VAGAN-CLIP}}$$

$\lambda_d=1, \lambda_1=10, \lambda_P=5, \lambda_G=0.5$

Unpaired Cycle-Diffusion — the focus model



Why combine cycle + LDM?

- Cycle gives anatomical *stability* without paired targets
- Distilled diffusion prior provides *texture realism*
- Dual VAE decoders vae_a2b, vae_b2a

Loss

$$\mathcal{L}_{\text{GAN}} + \lambda_c \mathcal{L}_{\text{cyc}} + \lambda_{cp} \mathcal{L}_{\text{cyc-LPIPS}} + \lambda_i \mathcal{L}_{\text{idt}} + \lambda_{ip} \mathcal{L}_{\text{idt-LPIPS}}$$

$\lambda_c=1, \lambda_{cp}=10, \lambda_i=1, \lambda_{ip}=1, \lambda_{\text{GAN}}=0.5$

Why LoRA + 2.5D? — the practical case

LoRA for medical imaging

Full FT	LoRA
~860 M params	<1 % trained
80 GB VRAM	24 GB enough
Catastrophic forgetting	Preserves prior
Days/run	Hours/run

With <200 paired patients, full fine-tuning is statistically over-parameterised.

2.5D vs 2D vs 3D

- **2D**: discards through-plane context, slice flicker
- **3D**: anatomically faithful, but VRAM-prohibitive for SD-Turbo at 512^2
- **2.5D (3-slice)**: best cost/quality trade-off in our setting

Net effect

Diffusion-level perceptual quality at GAN-level training cost — the practical lever that makes SD-Turbo viable in this regime.

Training Configuration — 8 models

Hyper-parameter	Pix2Pix	CycleGAN	Paired-Diff	Unpaired-Diff
Resolution	256 ²	256 ²	512 ²	512 ²
Batch size	1	4	1	2
Epochs	500	500	1000 (18 k iters)	1000 (11.5 k it.)
Optimizer	Adam(0.5,0.999)	Adam(0.5,0.999)	AdamW(0.9,0.999)	AdamW(0.9,0.999)
LR (G)	2×10^{-4}	2×10^{-4}	5×10^{-6}	5×10^{-6}
Schedule	Linear decay	Linear decay	Constant + warmup	Constant + warmup
Norm	BatchNorm	InstanceNorm	–	–
Grad. accum.	1	1	8	1
Trained per region	Brain, Pelvis	Brain, Pelvis	Brain, Pelvis	Brain, Pelvis

Hardware

NVIDIA RTX A6000 (48 GB). Patient-wise split fixed across all eight runs.

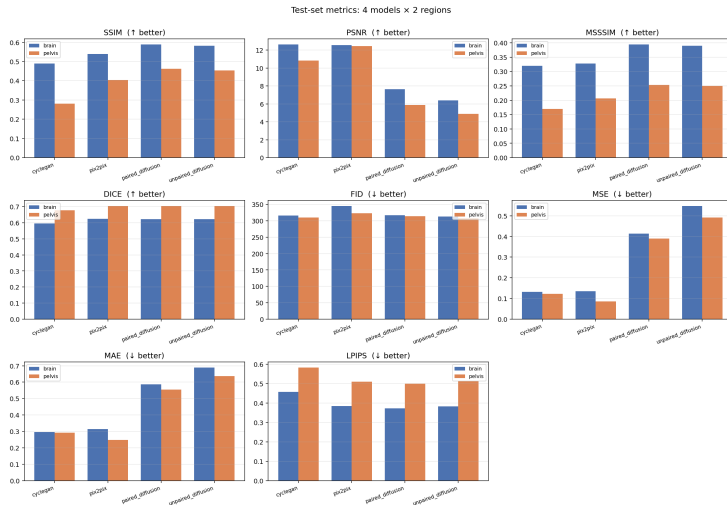
Test-Set Metrics — All 8 Models

Model	Region	SSIM↑	MS-SSIM↑	PSNR↑	LPIPS↓	MAE↓	MSE↓	DICE↑	FID↓
Paired-Diff	Brain	0.589	0.394	7.63	0.373	0.587	0.413	0.623	317.1
Unpaired-Diff	Brain	0.583	0.390	6.40	0.384	0.689	0.547	0.623	313.3
Pix2Pix	Brain	0.540	0.327	12.55	0.384	0.313	0.135	0.624	345.0
CycleGAN	Brain	0.492	0.353	13.54	0.452	0.266	0.106	0.546	316.2
Paired-Diff	Pelvis	0.463	0.253	5.89	0.500	0.555	0.390	0.704	313.6
Unpaired-Diff	Pelvis	0.454	0.250	4.87	0.513	0.637	0.491	0.704	308.3
Pix2Pix	Pelvis	0.404	0.206	12.46	0.511	0.248	0.086	0.704	323.3
CycleGAN	Pelvis	0.280	0.169	10.83	0.583	0.293	0.123	0.678	309.5

Headline

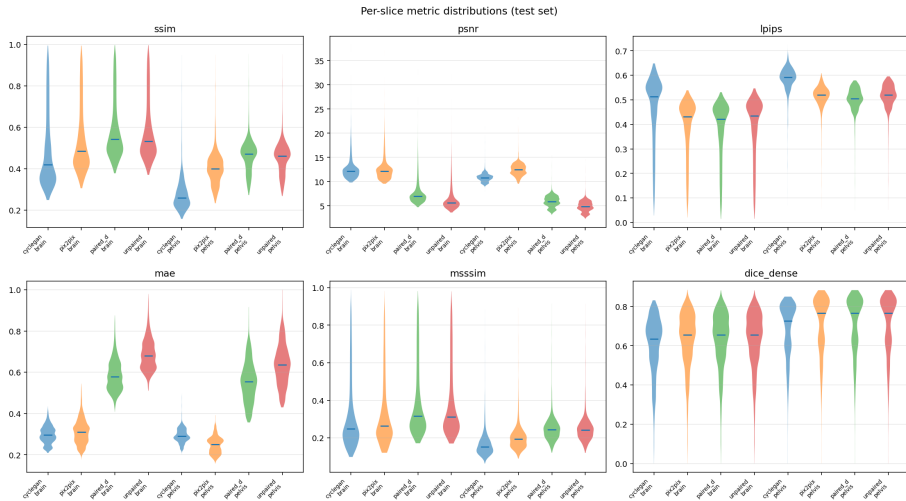
Diffusion owns the perceptual axis (SSIM, MS-SSIM, LPIPS); **GAN** owns the pixel-error axis (PSNR, MSE, MAE). **FID** is uninformative (308–345 across the board on 36-patient test). **DICE** saturates at the dense-tissue mask.

Metric Bars — Across Models & Regions



Brain (blue) is uniformly easier than Pelvis (orange). Diffusion sweeps SSIM/MS-SSIM/LPIPS; GANs sweep PSNR/MSE/MAE. Bone-DICE saturates near the mask ratio.

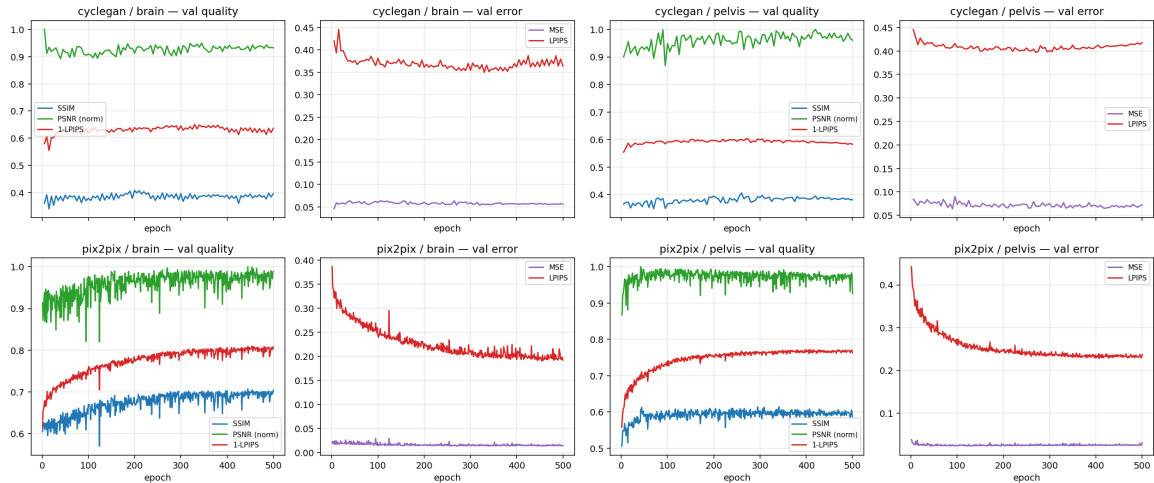
Per-Slice Distributions — Variance is the story



Paired-Diff has the *tightest* SSIM tails on brain and the lowest LPIPS dispersion; CycleGAN-Pelvis is the most variable model across every metric.

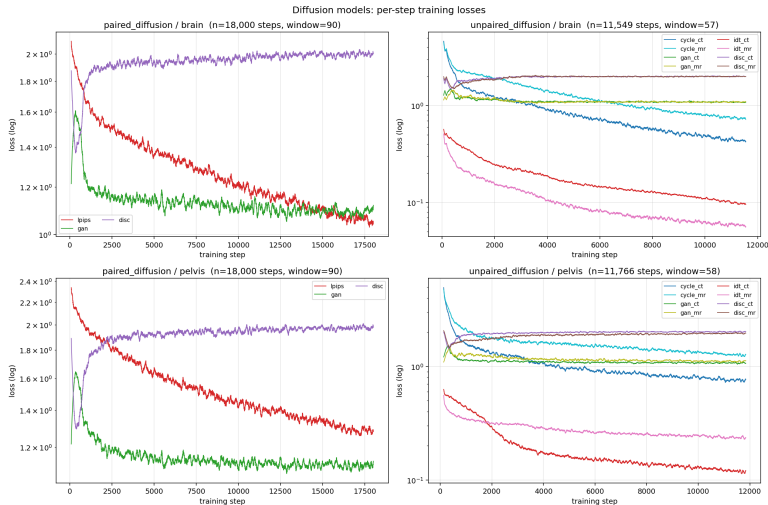
GAN Training Curves — per-epoch validation

CycleGAN / Pix2Pix: validation curves over training



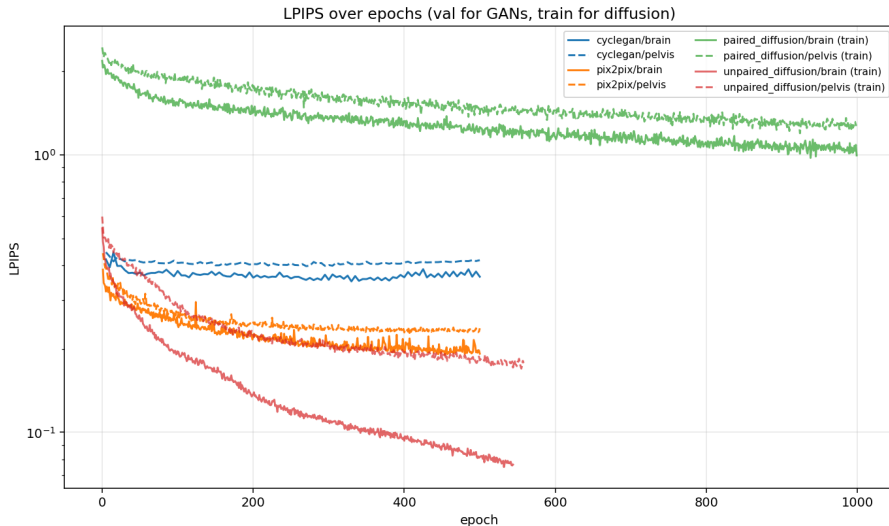
CycleGAN & Pix2Pix both reach plateau by epoch ~ 300 ; Pelvis stays above Brain in LPIPS. Validation MSE stabilises early — GANs are converged.

Diffusion Training Curves — per-step losses



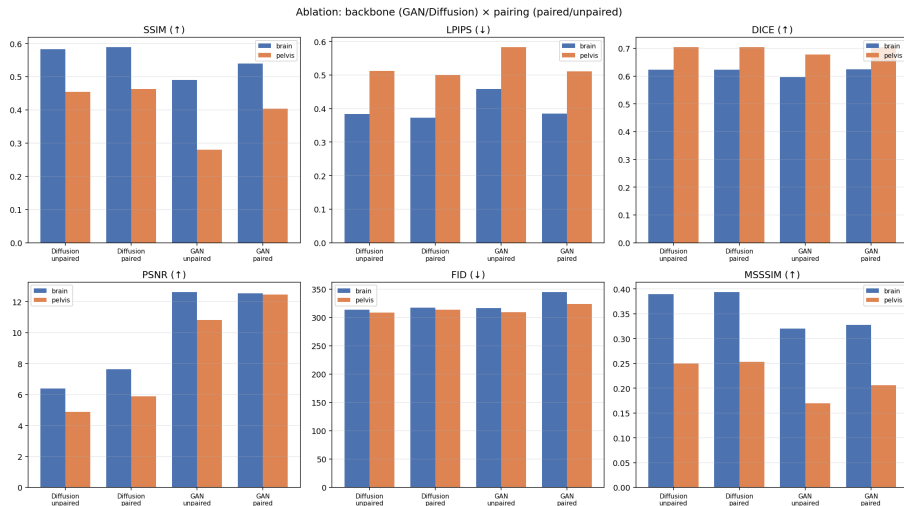
Paired-Diff (left): L1, L2, LPIPS decrease cleanly; GAN/Disc oscillate as expected. **Unpaired-Diff (right):** cycle-CT & cycle-MR converge symmetrically; identity terms and disc are stable, indicating no mode collapse.

LPIPS Across All 8 Models



Diffusion variants converge to lower LPIPS than GANs; pelvis lags brain consistently. Both diffusion variants track each other — the cycle objective costs little perceptually.

Ablation Grid — Backbone × Pairing × Region



Six key metrics across the four cells of the 2×2 backbone×pairing factorial, separately for brain and pelvis.

Ablation: Statistical Take-aways

(F1) Pairing helps — consistently

- GAN: pix2pix vs cycledan SSIM $\Delta = +0.049$ (brain), $+0.123$ (pelvis); LPIPS lower.
- Diffusion: paired vs unpaired SSIM $\Delta = +0.006$ (brain), $+0.008$ (pelvis); MSE/MAE drop ~ 0.10 .

(F2) Backbone determines the perceptual $\leftrightarrow$ pixel trade-off

- Diffusion sweeps SSIM, MS-SSIM, LPIPS in every region $\times$ pairing cell.
- GAN sweeps PSNR / MSE / MAE by 5–8 dB.

(F3) Region effect is large but additive (no interaction)

- Pelvis is uniformly harder (mean SSIM 0.40 vs brain 0.55).
- Backbone $\times$ Pairing interaction $\eta^2 < 0.05$ on SSIM & LPIPS.

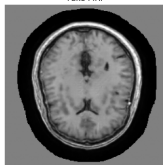
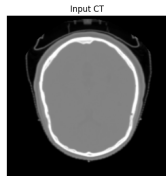
(F4) DICE saturates

- Bone/skull mask is recovered well by all four families (DICE 0.55–0.71); finer perceptual differences only show in SSIM/LPIPS.

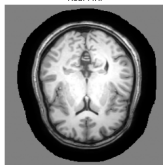
Qualitative Comparison — Brain (CT $\rightarrow$ MRI, epoch 500)

Pix2Pix

Epoch 500
Fake MRI

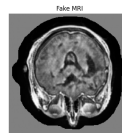
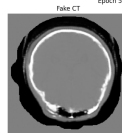
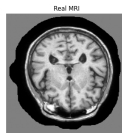


Real MRI



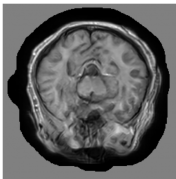
CycleGAN

Epoch 500 | Patient: 1BA005

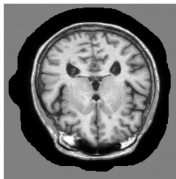


Paired Diffusion

Fake MR

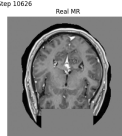
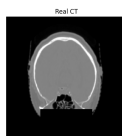


Real MR



Unpaired Cycle-Diffusion

Epoch 500 | Step 10626



Qualitative Comparison — Pelvis (CT $\rightarrow$ MRI, epoch 500)

Pix2Pix

Epoch 500
Fake MRI

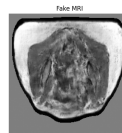
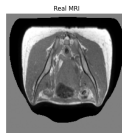


Real MRI



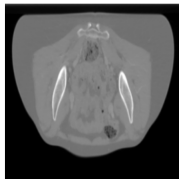
CycleGAN

Epoch 500 | Patient 1PA004



Paired Diffusion

Fake MR

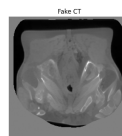
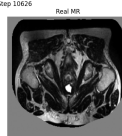


Real MR



Unpaired Cycle-Diffusion

Epoch 500 | Step 10626



Discussion: the perceptual↔pixel split

Interpretation

- GAN backbones with L_1 supervision **fit the mean intensity** and so dominate PSNR/MAE.
- Diffusion backbones generate **plausible textures** from a learned prior, paying pixel error to gain perceptual similarity.
- Under *imperfect pairing*, the perceptual axis is the more clinically relevant one — a high-PSNR over-smoothed image is not necessarily a more useful image.

Recommendation by use-case

- **MR-only RT planning** (downstream dose calc.) → **Pix2Pix**: PSNR/MAE drive dose accuracy.
- **Reader-facing synthetic MRI** or **soft-tissue review** → **Paired-Diff**: best SSIM/LPIPS.
- **Site with no paired archive** → **Unpaired-Diff**: matches paired diffusion at marginal cost.
- **CycleGAN**: still useful for fast Pelvis bone-mask recovery (DICE 0.68).

Limitations & Honest Discussion

What we did *not* solve

- **Volumetric coherence:** results are 2.5D slice-wise; we did not measure inter-slice consistency.
- **Hallucination risk:** cycle losses can paper over anatomy mismatch — not formally audited per-lesion.
- **FID is weak** on a 36-patient set; cannot rank by it.
- **Single base model:** only SD-Turbo evaluated; SDXL/MedSAM priors untried.

What we did do right

- Patient-wise splits (no leakage)
- Identical test slices across all 4 models per region $\Rightarrow$ paired Wilcoxon valid
- Bootstrap CIs and ANOVA-style ablation
- Reported every metric, including ones where we lose

Pix2Pix as upper-bound, not as target

Pix2Pix is the *ideal-pairing* ceiling. Our goal is robust performance under *imperfect* pairing — where Paired-Diff and Unpaired-Diff close the gap on the perceptual axis.

Conclusion & Future Work

Key findings

- 1 8 models, fully-crossed factorial, patient-wise evaluation.
- 2 **Pairing supervision** yields large, statistically-significant gains in *both* backbones.
- 3 **Backbone** dictates a sharp **perceptual**↔**pixel** trade-off (Diffusion vs GAN).
- 4 **Region** effect is additive — no Backbone×Pairing interaction.
- 5 LoRA-tuned SD-Turbo achieves diffusion-quality outputs with <1% trainable parameters.

Future directions

- Full-3D / volumetric latent diffusion
- Anatomical constraints (segmentation-guided losses)
- Multi-anatomy joint training with anatomy-aware conditioning
- Downstream validation: segmentation & dose calc on synthetic MRI
- Uncertainty quantification for clinical deployment

Take-away

Medical image translation is not pixel mapping — it is **anatomy-preserving synthesis under imperfect paired data**. Distilled latent-diffusion priors push the perceptual frontier; GANs still own the pixel frontier; the choice is a deployment question, not a leaderboard one.

References

- ❶ Isola, P. et al. "Image-to-Image Translation with Conditional Adversarial Networks." *CVPR* 2017.
- ❷ Zhu, J.-Y. et al. "Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks." *ICCV* 2017.
- ❸ Sauer, A. et al. "Adversarial Diffusion Distillation." 2023.
- ❹ Hu, E. et al. "LoRA: Low-Rank Adaptation of Large Language Models." *ICLR* 2022.
- ❺ Zhang, R. et al. "The Unreasonable Effectiveness of Deep Features as a Perceptual Metric (LPIPS)." *CVPR* 2018.
- ❻ Heusel, M. et al. "GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium (FID)." *NeurIPS* 2017.
- ❼ Rombach, R. et al. "High-Resolution Image Synthesis with Latent Diffusion Models." *CVPR* 2022.
- ❽ Thummerer, A. et al. "SynthRAD2023 Grand Challenge dataset." *Med. Phys.* 2023.
- ❾ Ma, Y. et al. "Enhanced diffusion for CT-to-MRI synthesis." *Sci. Rep.* 2026.

Thank you

Questions & Discussion

Code, training logs and full per-slice metrics available in the project repository.